

Hey, Central Limit Theorem, why that shape?

Ishan Goel

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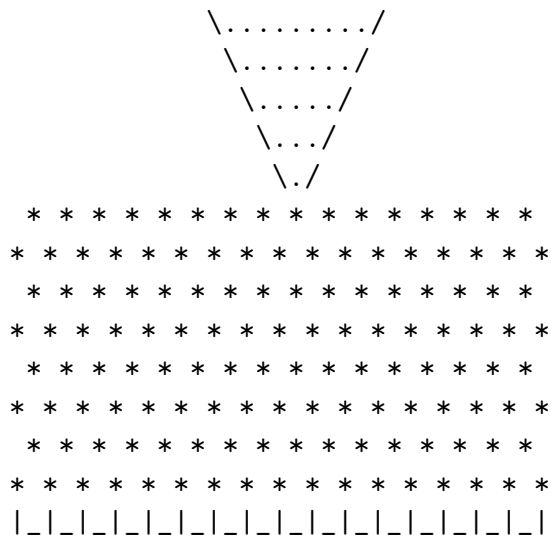
Fundamental to statistics and thus our ability to do science, the Central Limit Theorem gives a remarkable guarantee: that the average of many independent samples from *any* distribution itself approaches one specific distribution. This special distribution has the shape of $\exp(-x^2)$. This document strives to somewhat informally give intuition for why the CLT approaches this shape.

Having used the CLT many times, most recently to make a very high speed FPGA-based Gaussian noise generator, I felt I needed to have more intuition for why this one specific shape is special.

We are lucky that in our universe we have a fat-humped tiny-tailed shape as the one CLT approaches, for without it, gaining certainty through sampling — and thus science — would be much harder.

1 Why bell shape?

First let's think about a fun physics experiment / toy. Imagine you had 100 tiny metal balls you could drop into the center of this grid of pegs.



As each ball falls into one of the bins, it hits a bunch of pegs. At each peg, the ball is randomly bumped either one step to the right or one step to the left.

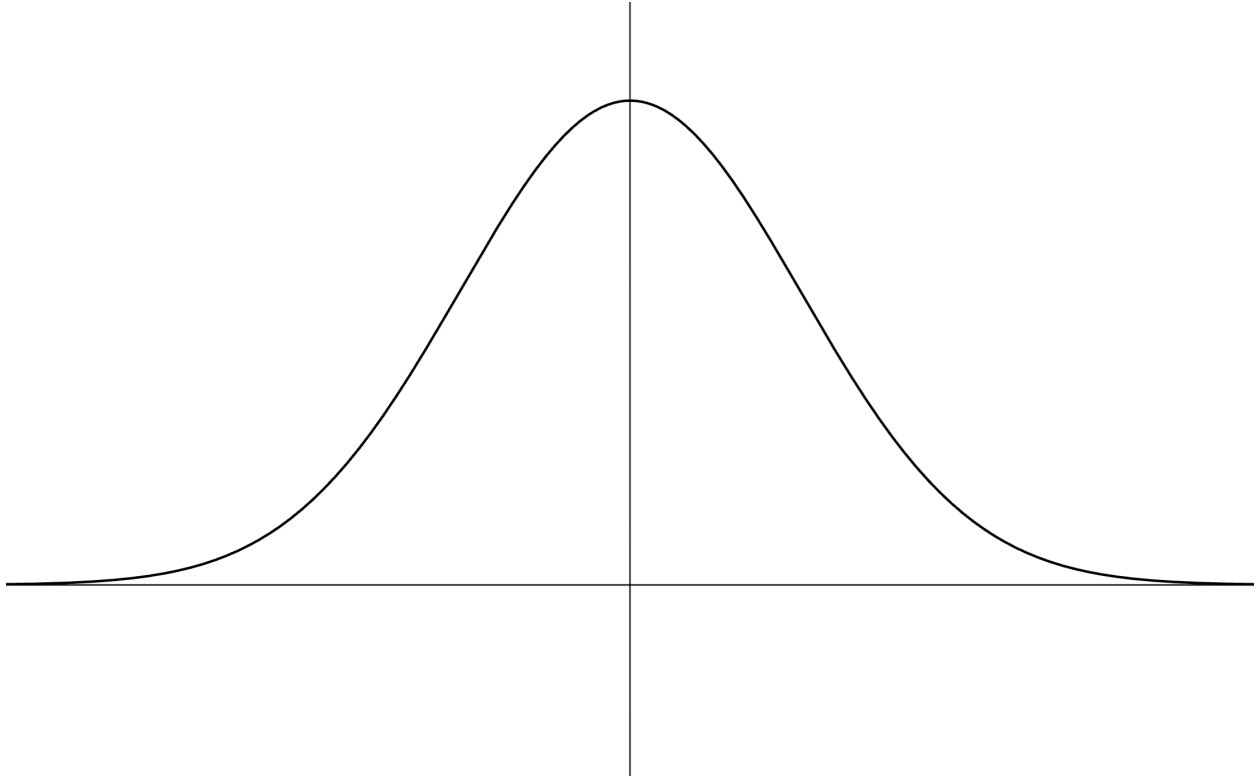
How many balls do you expect in each bin?

If you imagine each peg as changing the position of the ball by $+1$ or -1 , it's easy to see that it'll be rare for the balls to fall far from the center, since that requires getting a big string of $+1$, $+1$... or a big string of -1 , -1 , ...

Meanwhile, there's a lot of ways to get to the center. As long as there's an equal number of $+1$ s and -1 s, the ball goes to the center.

You can imagine the final position of each ball to be a random variable X that is the sum of a bunch of random binary choices $\{-1, +1\}$. Having dropped 100 balls into our setup, by counting the balls in each bin, we can figure out the chance of X taking on a certain value.

Now we can finally see the connection to the Central Limit Theorem. Having summed a bunch of samples from the distribution $\{-1, +1\}$ we got a resulting distribution that must be fat in the middle and thin in the tails: some sort of hump.



It's easy to convince yourself of this. If you had 8 samples of -1 or $+1$, how many times would the sum be 0 vs 8 or -8 ? Can you try creating a general formula for the number of ways a certain sum can be obtained given n samples? What happens when you plot this formula?

2 Why $\exp(-x^2)$?

Well, okay, so of course we expect a bell shape. But why is the Gaussian, $\exp(-x^2)$, the special shape that everything converges to? This is the main content of this document and what I've been thinking about for a while. I've been able to break down the problem into a bunch of facts combined with some mild hand-waving. These are the best underlying intuitions I can offer. They are however still somewhat technical, and so it's hard to claim these as "intuitive."

Anyway, here's what we hope to make obvious:

1. Adding independent samples from two probability distributions results in a variable distributed as the convolution of those two distributions.

$$\begin{aligned} X &\sim f_X \\ Y &\sim f_Y \\ X + Y &\sim f_X * f_Y \end{aligned}$$

2. Convolution is multiplication in Fourier space.

$$\begin{aligned} \text{Let } \mathcal{F}f(x) &= F(k) \text{ and } \mathcal{F}g(x) = G(k) \\ C(k) &= F(k) \cdot G(k) \\ \text{Then } f * g &= \mathcal{F}^{-1}C(k) \end{aligned}$$

3. The Fourier transform of a Gaussian is a Gaussian.

$$ce^{-ax^2} \xrightarrow{\mathcal{F}} c'e^{-a'x^2}$$

4. Gaussians multiplied are still Gaussian.

$$ce^{-ax^2} \cdot c'e^{-a'x^2} = (c'c)e^{-(a'+a)x^2}$$

Maybe you can already tell what's happening here! Addition means convolution. And convolution means multiply in Fourier space. And Gaussians somehow go into Fourier space and multiply and go back while keeping their shape, so of course Gaussians are the magic shape.

3 Fact 1: Adding is convolution

We will start with an example that we will repeatedly generalize.

You have probably thrown two six-sided dice and added the results many times by now. How likely is it that the sum is 7? What about 12? Or 2? It's easy to see that there's many ways to get a 7, but only one way to get 12 or 2.

Specifically, (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1), all add to 7, and only (6, 6) or (1, 1) add up to 12 or 2, so getting a 7 is 6x more likely than getting a 12 or a 2. Of course, our argument here needs each value to occur with equal probability. Wanting to generalize as mathematicians do, you might wonder: what if the probabilities aren't all equal?

Let $P(n)$ = Probability of n appearing.

Let $S(n)$ = Probability of the sum being n

$$\begin{aligned} S(7) &= P(1)P(6) + P(2)P(5) + \dots \\ &= \sum_{i=1}^6 P(i)P(7-i) \end{aligned}$$

It's easy to see why this works. To generate the sum 7, we need to add up every way we can get the number i and $7-i$. What if we want a different sum?

$$\begin{aligned} S(5) &= P(1)P(4) + \dots \\ &= \sum_{i=1}^6 P(i)P(5-i) \text{ }^1 \\ S(n) &= \sum_{i=1}^6 P(i)P(n-i) \end{aligned}$$

Now, generalizing to a die of any number of sides, we get a formula for the distribution of the sum of two independent samples from a discrete distribution.

$$S(n) = \sum_i P(i)P(n-i)$$

Finally, we generalize to continuous probability distributions and recognize this operation as something called convolution.

$$\begin{aligned} S(n) &= \int_{-\infty}^{\infty} P(x)P(n-x) dx \\ &= (P * P)(n) \end{aligned}$$

We now know how to get the distribution of the sum of two samples.

¹The astute reader may notice that the sum involves terms such as $P(0)$ and $P(-1)$. These are ok, since they can be defined as 0.

4 Fact 2: Convolution is multiplication

It is a standard fact that convolving two functions is equivalent to multiplying their Fourier transforms and inverse-transforming. I will justify this for completeness.

First, let's look at a direct proof.

$$\begin{aligned}
 (a * b)(t) &= \int_{-\infty}^{\infty} a(\tau)b(t - \tau) d\tau && \text{Definition of convolution} \\
 \mathcal{F}(a * b)(k) &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} a(\tau)b(t - \tau) d\tau \right] e^{-2\pi i k t} dt && \text{Fourier transform} \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} a(\tau)b(t - \tau)e^{-2\pi i k t} dt d\tau^2 \\
 &= \int_{-\infty}^{\infty} a(\tau) \int_{-\infty}^{\infty} b(t - \tau)e^{-2\pi i k t} dt d\tau && f(\tau) \text{ independent} \\
 &= \int_{-\infty}^{\infty} a(\tau) \int_{-\infty}^{\infty} b(u)e^{-2\pi i k(u+\tau)} du d\tau && \text{u-substitution, } u = t - \tau \\
 &= \int_{-\infty}^{\infty} a(\tau)e^{-2\pi i k \tau} \left(\int_{-\infty}^{\infty} b(u)e^{-2\pi i k u} du \right) d\tau \\
 &= \left(\int_{-\infty}^{\infty} a(\tau)e^{-2\pi i k \tau} d\tau \right) \left(\int_{-\infty}^{\infty} b(u)e^{-2\pi i k u} du \right) && \text{Independent subintegral} \\
 \mathcal{F}(a * b) &= \mathcal{F}(a) \cdot \mathcal{F}(b) && \text{Tada!}
 \end{aligned}$$

If you're like me, this is not terribly satisfying. You probably want to know the *essence* of why Fourier transforms and convolutions have anything to do with each other. The following is for this reader.

²Worried about the order of integration? Look up Fubini's theorem and think about why it applies.

The essence of the Fourier transform is the idea that functions can be represented as a weighted sum of a bunch of waves: terms looking like $e^{2\pi i k x}$, where k is the wave's frequency:

$$\begin{aligned} a(x) &= A_0 e^{2\pi i \cdot 0 x} + A_1 e^{2\pi i \cdot 1 x} + A_2 e^{2\pi i \cdot 2 x} + \dots \\ b(x) &= B_0 e^{2\pi i \cdot 0 x} + B_1 e^{2\pi i \cdot 1 x} + B_2 e^{2\pi i \cdot 2 x} + \dots \end{aligned}$$

What if we were to multiply $a(x)$ and $b(x)$?

$$c(x) = a(x) \cdot b(x) = (A_0 B_0) e^{2\pi i \cdot 0 x} + (A_0 B_1 + A_1 B_0) e^{2\pi i \cdot 1 x} + (A_0 B_2 + A_1 B_1 + A_2 B_0) e^{2\pi i \cdot 2 x} + \dots$$

Looking familiar yet? It seems that each coefficient is the sum of all the coefficients with indices adding to the right number...

Of course! To find out the 'weight' on a term like $e^{2\pi i k x}$ we must use every possible combination of coefficients that sum to k .

Letting $F(k)$ denote the function returning the k -th coefficient³ in the expansion for $f(x)$, we see that:

$$C(k) = \sum_i A(i) B(k - i)$$

We can recognize this as convolution.

See what's happened? Expressing two functions as weighted sums of waves, we found that multiplying the functions results in a convolution of the coefficients.

Finally, we recognize that in the continuous domain, $F(k)$ becomes the Fourier transform, and so:

$$\begin{aligned} C(k) &= A(k) * B(k) \\ \mathcal{F}(a \cdot b) &= \mathcal{F}(a) * \mathcal{F}(b) \end{aligned}$$

Tada! We've intuited the relation between the Fourier transform and convolution.

However! The careful reader may notice that we have proved a slightly different result from earlier. Our integrals proved:

$$\mathcal{F}(a * b) = \mathcal{F}(a) \cdot \mathcal{F}(b)$$

Well, which is it? It turns out that both are true, consistent, and can be derived from each other (challenge: try using the fact that $\mathcal{F}\mathcal{F}f(x) = f(-x)$ and that $\mathcal{F}^3 = \mathcal{F}^{-1}$ to demonstrate this).

It seems that the Fourier transform provides us a great freedom: to change convolution into multiplication, and to change multiplication into convolution. We will soon exploit this.

³ k -th in the sense that it is the coefficient of the $e^{2\pi i k x}$ term

5 Fact 3: Gaussian $\xrightarrow{\mathcal{F}}$ Gaussian

To recap where we are right now, remember that we've shown that adding random variables is the same as convolving their probability distributions, and that we can turn convolution into multiplication via the Fourier transform. Now, if we're able to show that Gaussians can go through the Fourier transform and back with shape maintained, we will be closer to our goal. Let's do it.

$$\begin{aligned} g(x) &= e^{-ax^2} && \text{Gaussian} \\ \mathcal{F}(g)(k) &= \int_{-\infty}^{\infty} e^{-ax^2} e^{-2\pi i k x} dx && \text{Fourier transform} \end{aligned}$$

If you try solving that integral, you will quickly find it to become quite complicated, and that ideas from fields like 'complex analysis' may be needed. As amateur mathematicians, we will instead resort to humble, yet powerful tricks such as viewing the Fourier transform as a sum.

We will prove some properties of the Fourier transform that apply to all functions, and use them to show that a Gaussian must be the Fourier transform of a Gaussian.

First, let's establish some subscript notation that will help us write illuminating sums.

$$\text{Let } F_k = F(k) = \mathcal{F}f(x)$$

With this notation, we're leaning into interpreting the Fourier transform as giving us a coefficient F_k at every frequency k (the Fourier transform tells us how strong each frequency is). Similarly, we will lean into seeing the function as a list of values indexed by x :

$$\text{Let } f_x = f(x)$$

Being fluid with our notation here allows us to choose when we want to give off 'function' aura or 'coefficient' aura. Don't stress about it! We will also use $\underline{e} = e^{2\pi}$ as shorthand.

Now we can write what the Fourier transform is all about: that a function is a weighted sum of waves.

$$f(x) = \dots + F_{-1}\underline{e}^{i(-1x)} + F_0\underline{e}^{i(0x)} + F_1\underline{e}^{i(1x)} + F_2\underline{e}^{i(2x)} + \dots = \sum_k F_k e^{2\pi i(kx)} \quad (1)$$

And how do we get those weights?

$$F(k) = \dots + f_{-1}\underline{e}^{-i(-1k)} + f_0\underline{e}^{-i(0k)} + f_1\underline{e}^{-i(1k)} + f_2\underline{e}^{-i(2k)} + \dots = \sum_x f_x e^{-2\pi i(kx)} \quad (2)$$

In a sense, the Fourier transform is thus a dot product of the function with waves of the form $e^{-2\pi i(kx)}$. Interpreting each dot product as measuring 'how much k -frequency is contained in

f ,⁴ it makes sense that, in the inverse transform, attaching the resulting dot products to the waves $e^{2\pi i(kx)}$ reconstructs the original function.

Again, we are being pretty fluid, and the equalities above are somewhat informal. Remember that these sums in reality are integrals and include all real numbers for k and x . We use sums since they feel easier to manipulate.⁴

To recap, here's what we have so far:

$$\begin{aligned} f(x) &= \sum_k F_k e^{2\pi i(kx)} \\ \text{Function} &= \sum_k (\text{Transform}) e^{2\pi i(kx)} \\ &\text{and} \\ F(k) &= \sum_x f_x e^{-2\pi i(kx)} \\ \text{Transform} &= \sum_x (\text{Function}) e^{-2\pi i(kx)} \end{aligned}$$

Considering how exponentials act nicely with differentiation, and seeing how both the Gaussian and the Fourier terms are densely populated with exponentials, you may try differentiating the sums to find some new rules.

$$\begin{aligned} f'(x) &= \sum_k F_k (2\pi i k e^{2\pi i(kx)}) \\ F'(k) &= \sum_x f_x (-2\pi i x e^{-2\pi i(kx)}) \end{aligned}$$

Now we realize that we can absorb the new constants resulting from differentiation into the existing coefficients.

$$\begin{aligned} \underbrace{f'(x)}_{\text{Function}} &= \sum_k \underbrace{(2\pi i k F_k)}_{\text{Transform}} e^{2\pi i(kx)} \\ \underbrace{F'(k)}_{\text{Transform}} &= \sum_x \underbrace{(-2\pi i x f_x)}_{\text{Function}} e^{-2\pi i(kx)} \end{aligned}$$

Interpret this as the creation of a new set of coefficients, and notice that each equation gives us a new Fourier transform pair. This is a cool step, so I will carefully explain both new results.

⁴Worried about the legality of each manipulation? Find a way to justify each step in the rigorous integral setting.

Consider the first equation:

$$\underbrace{f'(x)}_{\text{Function}} = \sum_k \underbrace{(2\pi i k F_k)}_{\text{Transform}} e^{2\pi i(kx)}$$

Whatever coefficients on the waves reconstruct the function must be, by definition, the Fourier transform of that function! So, the Fourier transform of $f'(x)$ is $2\pi i k$ times F_k :

$$\boxed{\mathcal{F}f'(x) = 2\pi i \cdot k F(k)}$$

Thus, we get a simple relation between the Fourier transforms of a function and its derivative⁵. Similarly, we can interpret the other equation:

$$\underbrace{F'(k)}_{\text{Transform}} = \sum_x \underbrace{(-2\pi i x f_x)}_{\text{Function}} e^{-2\pi i(kx)}$$

Given some function, the way we calculate its Fourier transform is by putting it in this weighted sum. So, by definition, $F'(k)$ must be the Fourier transform of $(-2\pi i x f_x)$ and so:

$$F'(k) = \mathcal{F}(-2\pi i \cdot x f(x))$$

and:

$$\boxed{\mathcal{F}(x f(x)) = \frac{i}{2\pi} F'(k)}$$

Finally, we can use these two boxed identities on a Gaussian.

$$\begin{aligned} g(x) &= e^{-ax^2} \text{ with } a > 0 \\ g'(x) &= -2a x g(x) \end{aligned}$$

Now, taking the Fourier transform of both sides, and applying both our identities:

$$\begin{aligned} \mathcal{F}g'(x) &= -2a \mathcal{F}(xg(x)) \\ 2\pi i \cdot k G(k) &= -2a \frac{i}{2\pi} G'(k) \end{aligned}$$

We obtain this differential equation for $G(k)$, the Fourier transform of our Gaussian:

$$\boxed{G'(k) = -\frac{2\pi^2}{a} k G(k)}$$

It may not seem like we've gotten much closer, but actually we are pretty much there. Notice how similar our equation for the derivative of the Gaussian we started with:

$$g'(x) = -2a x g(x)$$

⁵This relation can be used to easily calculate derivatives via the Fourier transform. This has computational benefits sometimes.

is to the differential equation we have. This must mean the transform is a Gaussian, as expected. We realize if we have $A = \frac{\pi^2}{a}$,

$$G(k) = C \cdot e^{-Ak^2}$$

solves our equation⁶!

All that remains is to find C.

$$\begin{aligned} C &= G(0) = \int_{-\infty}^{\infty} e^{-2\pi i(0)x} e^{-ax^2} dx \\ &= \int_{-\infty}^{\infty} e^{-ax^2} dx \\ \text{Let } u &= \sqrt{a}x \implies dx = \frac{1}{\sqrt{a}} du \\ C &= \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-u^2} du \end{aligned}$$

Using the value of this standard integral

$$C = \sqrt{\frac{\pi}{a}}$$

Finally! We have the Fourier transform of our Gaussian:

$$\boxed{\mathcal{F}e^{-ax^2} = \sqrt{\frac{\pi}{a}} e^{-\frac{(\pi k)^2}{a}}}$$

So, indeed, the Fourier transform of a Gaussian is a Gaussian⁷.

We can additionally normalize the original Gaussian so that it can function as a probability distribution (its integral from $-\infty$ to ∞ must be 1).

$$\boxed{\mathcal{F}\sqrt{\frac{a}{\pi}}e^{-ax^2} = e^{-\frac{(\pi k)^2}{a}}}$$

⁶Verify this.

⁷Notice, additionally, that the standard deviation of the transformed Gaussian is inversely proportional to the original. This has connections to Heisenberg's Uncertainty Principle in Quantum Mechanics.

6 Gaussian + Gaussian

Here's what we've shown so far:

1. Adding independent samples results in a variable distributed as the convolution of the two original distributions.

$$\begin{aligned} X &\sim f_X \\ Y &\sim f_Y \\ X + Y &\sim f_X * f_Y \end{aligned}$$

2. Convolution is multiplication in Fourier space.

$$f * g = \mathcal{F}^{-1}[F(k) \cdot G(k)]$$

3. The Fourier transform of a Gaussian is a Gaussian.

$$\mathcal{F} \sqrt{\frac{a}{\pi}} e^{-ax^2} = e^{-\frac{(\pi k)^2}{a}}$$

4. Gaussians multiplied are still Gaussian.

$$ce^{-ax^2} \cdot c'e^{-a'x^2} = (c'c)e^{-(a'+a)x^2}$$

You can probably see that we can prove that samples of a Gaussian random variables added together must be Gaussian distributed. Let's do it.

$$\begin{aligned} X &\sim \sqrt{\frac{a}{\pi}} e^{-ax^2} \\ X + X &\sim f_{X+X} \\ f_{X+X} &= \left(\sqrt{\frac{a}{\pi}} e^{-ax^2} \right) * \left(\sqrt{\frac{a}{\pi}} e^{-ax^2} \right) \\ &= \mathcal{F}^{-1} \left[e^{-\frac{(\pi k)^2}{a}} \cdot e^{-\frac{(\pi k)^2}{a}} \right] \\ &= \mathcal{F}^{-1} \left[e^{-\frac{2(\pi k)^2}{a}} \right] \\ \mathcal{F}[f_{X+X}] &= e^{-\frac{(2\pi k)^2}{a}} \end{aligned}$$

Notice that $\mathcal{F} \left[\sqrt{\frac{2a}{\pi}} e^{-2ax^2} \right] = e^{-\frac{(\pi k)^2}{2a}}$ and so:

$$\boxed{f_{X+X} = \sqrt{\frac{2a}{\pi}} e^{-2ax^2}}$$

See how easy the Fourier transform made that?